Project Report: Sinner–Alcaraz on Bluesky: A Social Network & Sentiment Analysis of the 2025 US Open

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Abstract

This report explores how the tennis rivalry between **Jannik Sinner** and **Carlos Alcaraz** shaped the online discourse on the **Bluesky** social platform during the **2025 US Open**. Gathering a dataset of **9,078 posts** via a day-by-day crawling strategy, we reconstruct the network of replies and mentions to map how communities formed, while tracing the emotional and sentiment dynamics triggered by match outcomes. The analysis reveals a highly modular network where fans cluster around shared news hubs rather than forming isolated echo chambers. Furthermore, by integrating neural sentiment signals with user interaction frequencies, we demonstrate how a sentiment-aware stance classifier can successfully refine and correct the classification errors inherent in simpler, frequency-only baselines.

Contents

1 Research Question

The goal of this project is to **characterise the online discourse** surrounding two of tennis's biggest rivals—Jannik Sinner and Carlos Alcaraz—on the Bluesky social platform during the 2025 US Open. In doing so, this study is guided by the following overarching **research question**:

"How do the Jannik Sinner and Carlos Alcaraz fan communities on Bluesky manifest at both a structural-topological and emotional level during a Grand Slam tournament like the US Open?"

To address this research question, we build a modular, reproducible NLP and social-network analysis pipeline. Concretely, the analysis aims to:

- **Describe** the social-network structure built from replies and mentions, identify cohesive communities, and measure user influence via centrality metrics.
- **Quantify** sentiment and emotion expressed in posts using state-of-the-art NLP models (RoBERTa, NRC lexicon, GoEmotions) and compare two emotion backends.
- **Assign a stance** (pro-Sinner, pro-Alcaraz or neutral) to each user by fusing sentiment signals with mention frequency, and evaluate how this refined stance differs from a simpler frequency-only baseline.
- **Visualise** all results to support interpretability—network graphs, sentiment time series, emotion profiles and stance distributions.

2 Data

2.1 Collection strategy

Data were collected from Bluesky using its official `atproto` API (`src/crawler.py`). Because the search endpoint does not guarantee exhaustive retrieval over a large window, a **day-by-day** crawling strategy was used:

- For each 24-hour window from `2025-08-21` to `2025-09-10` (UTC), two keyword queries were issued—one **Sinner-oriented** and one **Alcaraz-oriented**—each capped at **5,000 posts/day** to respect rate limits while capturing peak tournament activity.
- Results from both queries were merged and deduplicated by post URI.

2.2 Dataset description

After crawling and deduplication the dataset contained **9,078 posts**. The raw schema (`data/sinner_alcaraz_posts.csv`) is summarised in Table 1.

Field	Description
`created_at`	UTC timestamp
`uri`	Unique post identifier
`text`	Original post text
`author_handle`	User's handle
`author_did`	Decentralised identifier
`reply_count`, `like_count`, `repost_count`	Engagement metrics
`mentions`	List of mentioned user DIDs
`links`	URLs in the post
`reply_parent_did`	DID of the user being replied to
`query_source`	Which query found the post (sinner/alcaraz)

Table 1: Raw data fields collected from Bluesky.

2.3 Preprocessing

The raw CSV goes through a preprocessing phase to be transformed into an enriched dataset (`data/sinner_alcaraz_processed.csv`, 44 columns) that serves as the single source of truth for all downstream stages. The enrichment (`src/preprocessing.py`) includes:

- **Text cleaning** (HTML unescape, URL/handle removal, normalisation) and **lemmatisation** (spaCy + TweetTokenizer).
- **RoBERTa sentiment** via `cardiffnlp/twitter-roberta-base-sentiment-latest`, producing a `sentiment_category` (positive/neutral/negative) and a continuous `sentiment_compound` $= P(\text{pos}) - P(\text{neg})$.
- **Emotion** with two backends: the **NRC lexicon** (NRCLex) over the 8 Plutchik emotions, and **GoEmotions** (`SamLowe/roberta-base-go_emotions`), whose 28 fine-grained labels are collapsed onto the same 8 categories for comparison.
- **Named entity recognition** (spaCy) and **entity linking** to detect mentions of Sinner or Alcaraz.
- **Post- and user-level stance** (see Section 3.2).

Community vocabulary. A word cloud over the whole community (Figure 1) is dominated by the players' names and tournament/tennis vocabulary, confirming that the crawl captured on-topic discourse.

Figure 1: Word cloud of the community corpus—player names and US Open/tennis vocabulary dominate.

3 Methodology

The analytical pipeline is a strict DAG (orchestrated by `main.py`).

3.1 Network construction

Two graphs are built:

- an undirected graph G_u, where the edge weight is the number of interactions between users;
- a directed graph G_d that captures the flow of communication from an author to the exact user they are replying to or mentioning.

For both, we compute degree, closeness and betweenness centrality, in-/out-degree, and PageRank.

To understand how these users naturally cluster together, we apply the Louvain community detection algorithm to the undirected graph, while using the Infomap algorithm to detect communities within the directed graph.

Filtered subgraphs (we selected the top 5 largest communities) are used for visualisation and stance analysis.

3.2 Stance detection

Post-level stance assigns the RoBERTa compound score to the mentioned player for single-mention posts; dual-mention posts are split via sentence-level sentiment. Similarly, post-level emotion stance assigns the net GoEmotions (BERT) emotion valence (the sum of positive emotions: joy, trust, and anticipation, minus the sum of negative emotions:

anger, disgust, sadness, and fear) to the mentioned player. User-level net stance fuses mean sentiment, mention frequency, and mean emotion stance:

$$\text{net\_stance} = 0.50\,(\text{mean}_\text{s} - \text{mean}_\text{a}) + 0.15\,(\text{freq}_\text{s} - \text{freq}_\text{a}) + 0.35\,(\text{mean\_emo}_\text{s} - \text{mean\_emo}_\text{a})$$

Users are labelled `sinner` / `alcaraz` / `neutral` with a ± 0.05 threshold. Polarisation is measured via stance assortativity, the cross-stance edge ratio, and the standard deviation of net stance.

4 Results

4.1 Social Network Analysis

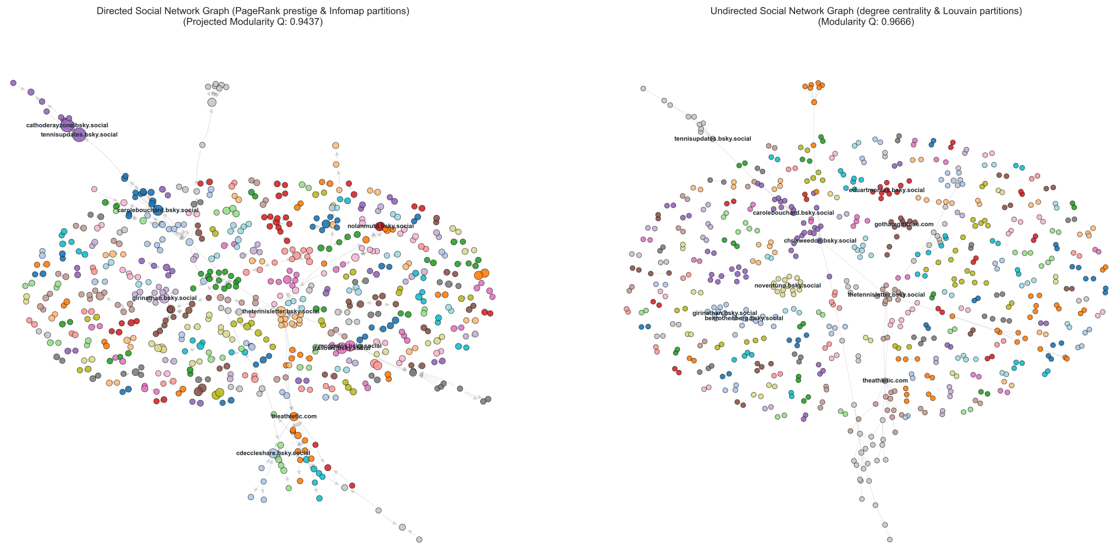
The interaction graph is **sparse but strongly modular**.

Out of 2,296 active users in the dataset, 683 engaged in interactions (replies or mentions) and form the nodes of our interaction network, while the remaining 1,613 users posted standalone content without direct interactive links. Global metrics (Table 2) computed on this interactive subset report a density of ≈ 0.002 for the undirected graph and ≈ 0.001 for the directed graph. Modularity scores are extremely high (Louvain ≈ 0.97, Infomap ≈ 0.94), suggesting communities cluster tightly around neutral-news-hub users rather than mixing. The giant connected component (GCC/WCC) spans only **111 users (≈16.3% of the interaction network)**, indicating that much of the interactive conversation happens in small, loosely linked clusters.

Metric	Undirected (G_u)	Directed (G_d)
Nodes	683	683
Edges	472	491
Density	0.00203	0.00105
Transitivity	0.02022	0.01282
Average clustering	0.01120	0.00798
GCC / WCC size	111	111
GCC / WCC fraction of users	16.25%	16.25%
SCC size	—	7
SCC fraction of users	—	1.03%
Louvain modularity	0.9666	—
Infomap modularity	—	0.9437

Table 2: Graph-level metrics for undirected (G_u) and directed (G_d) representations (`data/network_global_metrics.csv`).

We first compare the two community partitions: Louvain on the undirected graph G_u (modularity $Q = 0.967$) and Infomap on the directed graph G_d (projected modularity $Q = 0.944$). Although both scores are high, they describe the network very differently (Figure 2).

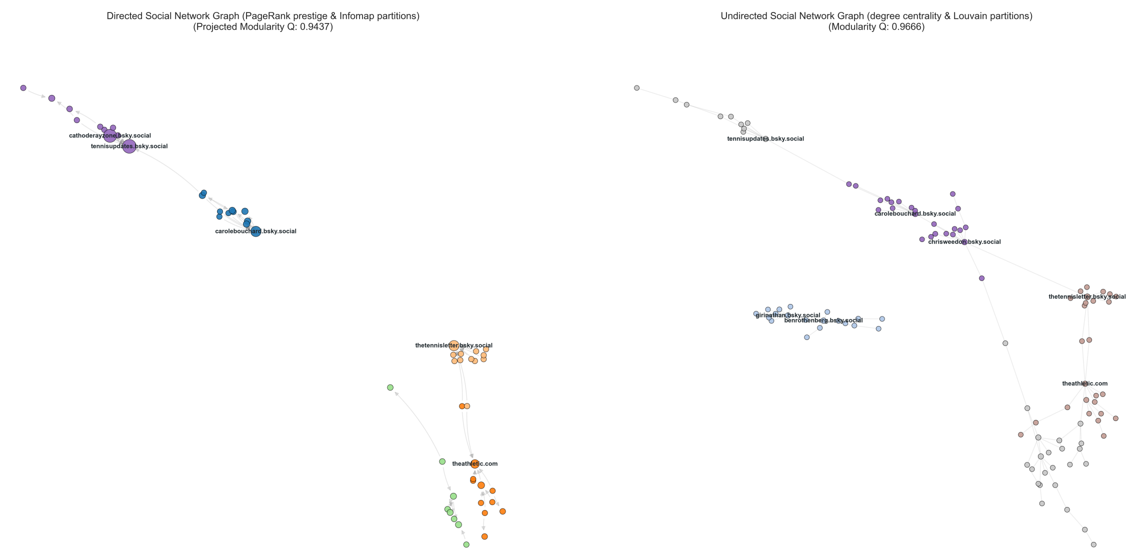


(a) Directed graph, Infomap partitions (node size ∝ PageRank prestige).

(b) Undirected graph, Louvain partitions (node size ∝ degree centrality).

Figure 2: Community-coloured social network: the directed/Infomap view (left) against the undirected/Louvain view (right).

The communities detected by Infomap on the directed graph offer a more granular and informative picture than the undirected approach. Whereas the undirected analysis (Louvain) tends to collapse the entire Giant Connected Component into a single, structurally dense block, the directed approach (Infomap) traces the *actual flow of information*. This makes it possible to identify functional sub-groups and bottlenecks that would otherwise remain invisible: where the undirected partition returns a single cohesive macro-community, the directed analysis reveals an articulated internal segmentation that better reflects the diffusion dynamics within the network. The same contrast holds on the filtered top-5 subgraphs (Figure 3), where the directed view separates the news hubs and their downstream chains far more cleanly.



(a) Filtered directed graph (Infomap). (b) Filtered undirected graph (Louvain).

Figure 3: Filtered top-5 communities: the directed view (left) exposes distinct hub-centred chains that the undirected view (right) blends together.

4.2 Social Sentiment Analysis

The RoBERTa classifier labels the corpus as overwhelmingly **neutral** (Figure 4): **77.2%** of posts are neutral, **16.2%** positive and only **6.6%** negative. The discourse is therefore strongly skewed towards informative, non-affective content, with positive posts outnumbering negative ones by roughly 2.5:1.

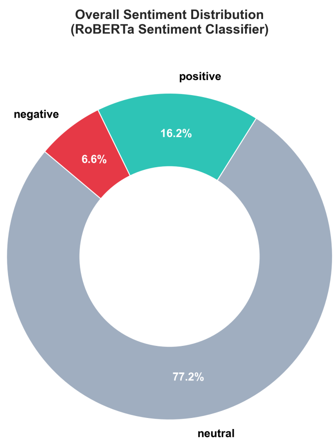
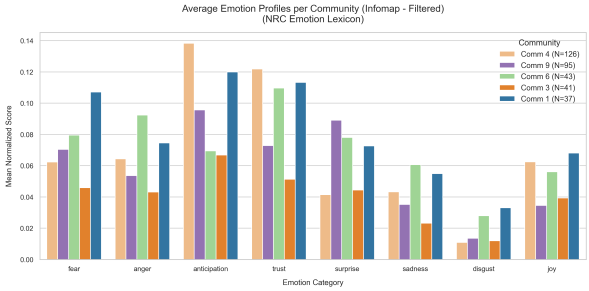


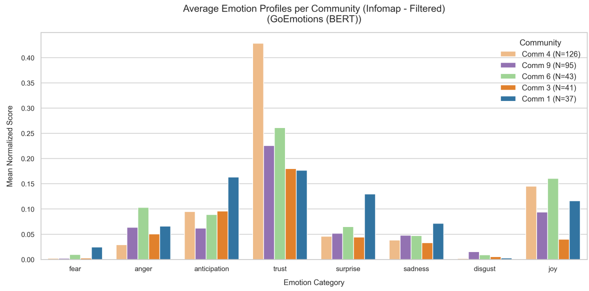
Figure 4: Overall sentiment distribution (RoBERTa): 77.2% neutral, 16.2% positive, 6.6% negative.

4.2.1 Emotion Analysis: NRCLex vs GoEmotions

Comparing the two emotion backends exposes a decisive advantage of the transformer model (Figure 5). NRCLex is a **bag-of-words** model: it looks up each word in an internal dictionary and checks which emotion the word is associated with. This causes posts mentioning **Sinner** to be classified under negative emotions (`fear`, `anger`, ...) because the lexicon interprets the word "sinner" as the religious sense of *sin*. **GoEmotions (BERT)**, by contrast, is a transformer that resolves *Sinner the tennis player* from *sinner the sin* from context. The effect is clearly visible in the dominant-emotion distribution (Figure 6): BERT assigns essentially **nobody** to `fear`, collapsing the spurious negative mass that NRC produces and reallocating it to `neutral` and `trust`.



(a) Per-community emotion profiles (NRC lexicon).



(b) Per-community emotion profiles (GoEmotions/BERT).

Figure 5: Average emotion profiles per Infomap community. NRC (left) inflates `fear`/`anger`; BERT (right) shifts mass towards `trust` and `anticipation`.

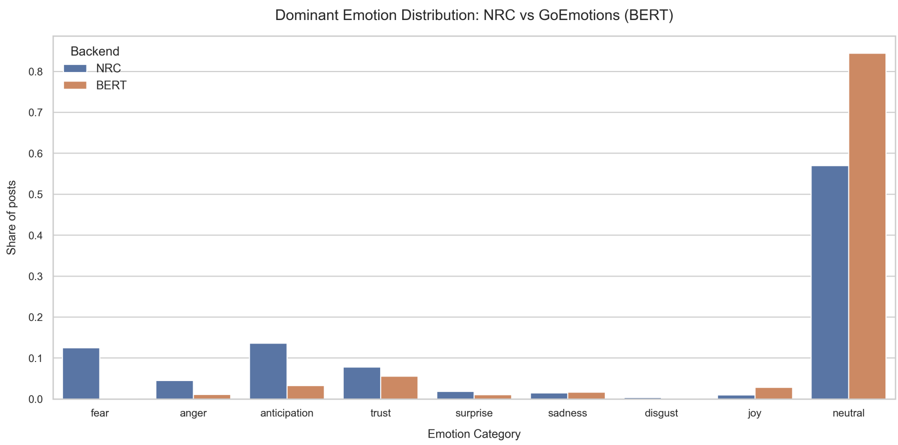


Figure 6: Dominant emotion distribution: NRC vs. GoEmotions (BERT). BERT removes the spurious `fear` peak induced by the "sinner"→"sin" lexical confusion and concentrates the distribution on `neutral`.

It is worth noting that both models operate **only in English**, so posts written in other languages are classified as neutral. This explains the high share of neutral posts observed in Section **??**, and points directly to a concrete avenue for future improvement (see Section 5).

The same BERT emotion signal can be projected back onto the directed network (Figure 7), colouring each node by its dominant emotion. The resulting map is dominated by neutral and trust-coloured nodes, consistent with the aggregate distribution, with localised pockets of anticipation and joy clustering around the most active hubs.

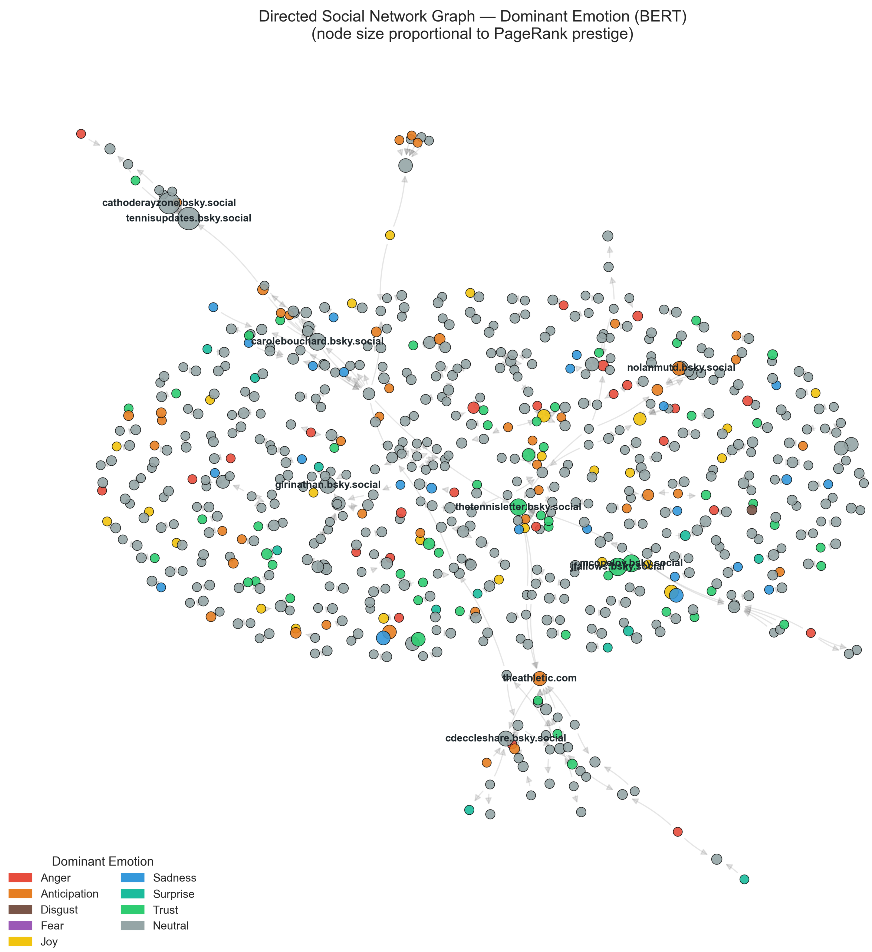


Figure 7: Directed social network coloured by dominant emotion (BERT); node size $\propto$ PageRank prestige.

4.3 Social Stance Analysis

Fusing sentiment, mention frequency and emotion into the user-level net stance (Section 3.2) yields the picture summarised in Figure 8. Of the **2,296** users analysed, **45.1%** (1,035) are neutral, **36.4%** (835) lean towards Alcaraz and **18.6%** (426) towards Sinner. *Alcaraz dominates the space*: his fanbase is nearly double the size of Sinner's.

From the *Top Communities – Mean Stance Score* panel we can see that the largest communities are predominantly neutral; only two opposing communities take a clear position, delineating one pro-Sinner community and one opposing pro-Alcaraz community.

Crucially, **no echo chambers exist**. Stance assortativity is slightly *negative* (-0.071) and **14.7%** of edges cross partisan lines (cross-stance edge ratio 0.147, net-stance standard deviation 0.327). Fans of opposing players actively interact with one another, bridged by shared, neutral sports-news hubs rather than retreating into isolated clusters.

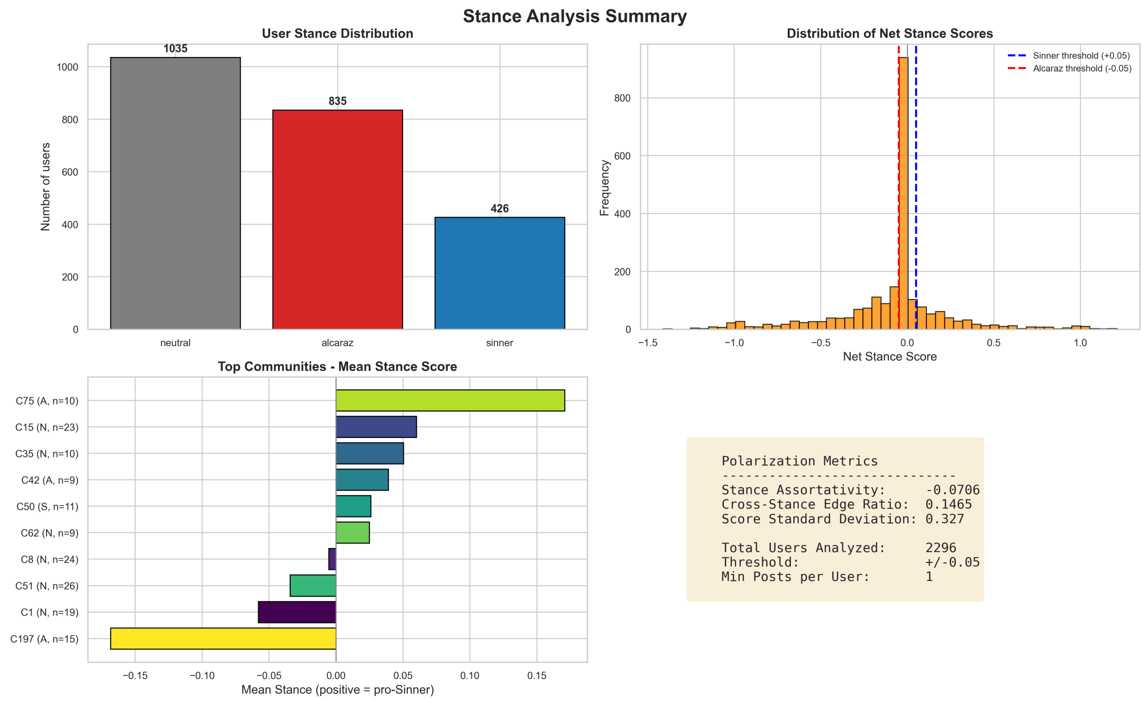


Figure 8: Stance analysis summary: user stance distribution, net-stance score distribution, per-community mean stance, and polarisation metrics.

Why mention frequency alone is not enough. Relying on mention frequency alone misclassifies users, because mentioning a player frequently says nothing about whether the user *supports* or *criticises* that player. Fusing sentiment with frequency is therefore methodologically essential to capture true support versus criticism.

Projecting the fused stance onto the directed graph (Figure 9) makes the spatial organisation of support explicit. Nodes coloured **orange** indicate a stance favourable to (or focused on) Sinner, while **blue** nodes represent an orientation towards Alcaraz. The two leanings are interleaved throughout the giant component rather than separated into distinct partisan regions, visually confirming the absence of echo chambers reported above.

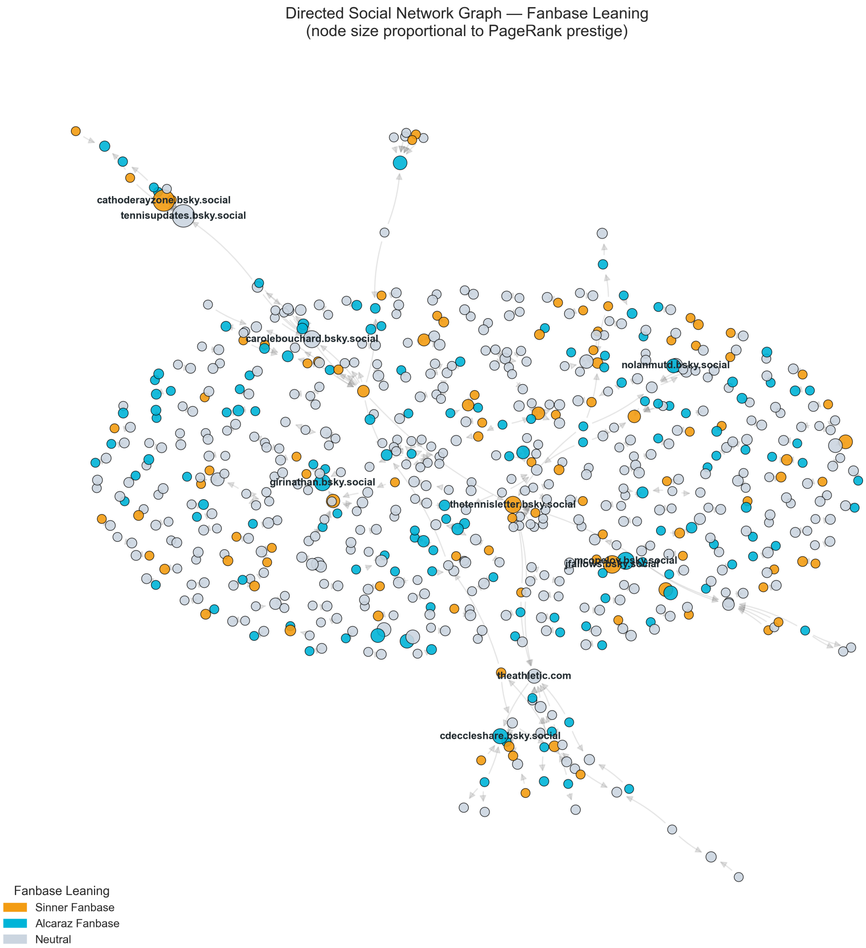


Figure 9: Directed social network coloured by fanbase leaning (orange = Sinner, blue = Alcaraz, light = neutral); node size $\propto$ PageRank prestige.

4.3.1 Sentiment dynamics over time

Tracking daily average sentiment against post volume across the tournament (Figure 10) reveals four clear patterns:

- **High event reactivity.** Online sentiment is purely reactive: match outcomes instantly shift the sentiment curves of both fanbases day by day.
- **The final divergence.** The final on **7 September** splits the emotional tone of the two communities: Alcaraz supporters reach their positive-sentiment peak (his victory), while Sinner supporters plunge towards a neutral-low score.
- **Civility under defeat.** Despite the disappointment of losing the final, the Sinner fanbase sentiment never drops deep into negative territory, indicating that disappointment did not turn into toxic behaviour or online hostility.
- **Late-tournament hype.** Engagement is highly concentrated: post volume grows exponentially during the final rounds (quarterfinals onwards), while the early rounds generate relatively low traction.

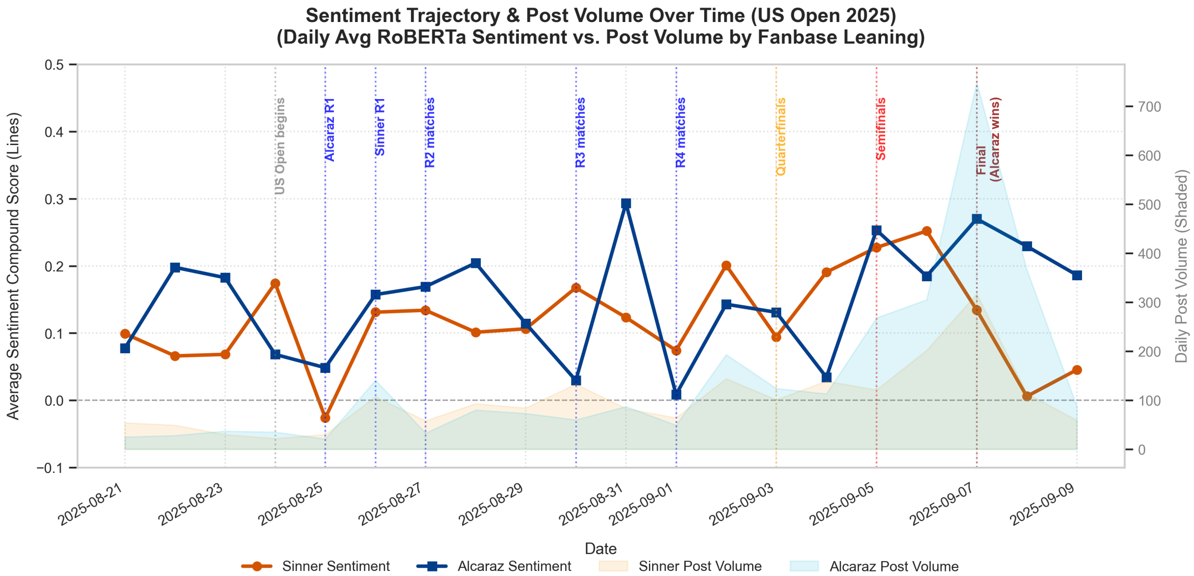


Figure 10: Sentiment trajectory and post volume over time (US Open 2025): daily average RoBERTa sentiment (lines) vs. post volume by fanbase leaning (shaded), annotated with tournament milestones.

5 Discussion

Taken together, the results answer the research question on both the structural-topological and the emotional level.

Structural-topological level. The Sinner and Alcaraz communities do *not* manifest as antagonistic echo chambers. The network is highly modular ($Q \approx 0.94$–0.97), yet that modularity reflects topical and hub-centred clustering rather than partisan segregation: stance assortativity is slightly negative (-0.071) and roughly one edge in seven crosses partisan lines (Section 4.3). The comparison between Louvain and Infomap (Section 4.1) is itself a finding: the choice of a *directed* model is what exposes the real information flow: where the undirected view sees a single dense macro-community, the directed view resolves an articulated internal structure of functional sub-groups bridged by neutral news hubs. These hubs, not partisan accounts, are the structural backbone that keeps the opposing fanbases connected.

Emotional level. Emotionally, the discourse is dominated by neutral and trust-laden content rather than hostility. Sentiment is strongly event-driven: it tracks match outcomes day by day and diverges sharply at the final, yet even in defeat the Sinner fanbase remains civil and never tips into sustained negativity (Section 4.3.1). The stance analysis adds nuance to this affective picture: Alcaraz commands the larger and more positively-charged fanbase, while a substantial neutral majority (45.1%) consumes and relays tournament news without taking sides.

Methodological contribution. Two methodological points stand out. First, **model choice matters for emotion**: the "sinner"→"sin" lexical trap shows that a bag-of-words backend (NRC) systematically fabricates `fear`/`anger` signal, whereas a context-aware transformer (GoEmotions/BERT) disambiguates the player's name and yields a far more faithful emotion distribution. Second, **stance requires more than counting mentions**: mention frequency alone conflates attention with allegiance, and only by fusing sentiment and emotion with frequency does the classifier recover genuine support versus criticism (Section 4.3).

Limitations and future work. The principal limitation is language coverage. Both emotion/sentiment models operate only in English, so non-English posts are pushed into the `neutral` class; this inflates the observed 77.2% neutral share and almost certainly understates genuine affect within the non-English portion of the community. Future improvements should therefore add language detection and multilingual sentiment/emotion models, which would sharpen both the temporal sentiment curves and the user-level stance scores. Further work could also extend the crawl window beyond a single tournament to test whether the absence of echo chambers persists outside the shared, high-salience context of a Grand Slam.

5.1 Conclusion

This study characterised the Sinner–Alcaraz discourse on Bluesky during the 2025 US Open by combining directed/undirected community detection, neural sentiment and emotion analysis, and a sentiment-aware stance classifier over a dataset of 9,078 posts. The picture that emerges is of a **highly modular but non-polarised** community: fans cluster around shared news hubs rather than partisan echo chambers, opposing fanbases interact across stance lines, and the emotional tone—while sharply reactive to match outcomes—stays predominantly neutral and civil, even in defeat. Methodologically, the work demonstrates that a context-aware transformer is necessary to avoid lexical artefacts in emotion detection, and that fusing sentiment with interaction frequency meaningfully refines and corrects a frequency-only stance baseline.